VIRAL DISEASES AT A GLANCE

Cutaneous lesions are an extremely common, and often the only, manifestation of many viral infections.

Viruses can produce skin lesions either by direct replication in the epidermis, as is the case with papillomaviruses, poxviruses, and several herpesviruses, or as a secondary manifestation of replication elsewhere in the body.

The direct effects of viral replication on epidermal cells and the inflammatory response of cell-mediated immunity combine to produce cutaneous lesions.

Neoplastic transformation or viral latency followed by reactivation occurs with certain viruses.

Laboratory diagnosis is by viral culture, microscopy, detection of viral nucleic acids or viral antigens, and/or serologic testing.

Many effective antiviral drugs exist, and others are currently in development.

Preventive techniques, most notably vaccines, have to date been the most effective means for controlling viral infection.

CLASSIFICATION

Viruses that infect animals are divided into several large families according to the size, shape, and structure of the virion and the type of viral nucleic acid within it, as shown in Table 191-1. Viruses of a given family can be identified based on these characteristics.

TABLE 191-1 Selected Animal Virus Groups

GROUP	SIZE (NM)	SYMMETRY	ENVELOPE	NUCLEIC ACID	CAPSID ASSEMBLY	EXAMPLES OF VIRUSES
Herpesvirus	120–200	Icosahedral	Yes	DNA	Nucleus	Herpes simplex virus (types 1 and 2), varicella-zoster virus, cytomegalovirus, EBV, KSHV/HHV-8
Papillomavirus	50–55	Icosahedral	No	DNA	Nucleus	Papillomaviruses

GROUP	SIZE (NM)	SYMMETRY	ENVELOPE	NUCLEIC ACID	CAPSID ASSEMBLY	EXAMPLES OF VIRUSES
Polyomavirus	45–50	Icosahedral	No	DNA	Nucleus	SV40
Poxvirus	240 × 300	Complex	No	DNA	Cytoplasm	Molluscum contagiosum, orf, milker's nodules virus, variola virus, vaccinia virus
Retrovirus	80–120	?	Yes	RNA	Cytoplasm	HIV, HTLV
Paramyxovirus	150–300	Helical	Yes	RNA	Cytoplasm	Measles virus, mumps virus
Togavirus	40–60	Icosahedral	Yes	RNA	Cytoplasm	Rubella virus